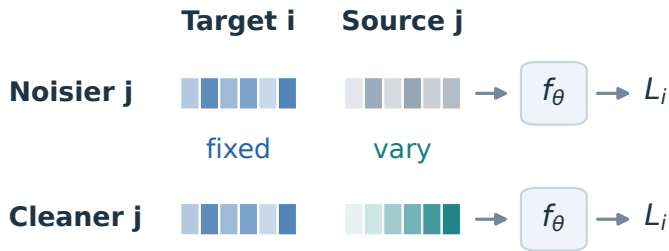


A. Controlled denoising

Fix target noise, other inputs, and model.



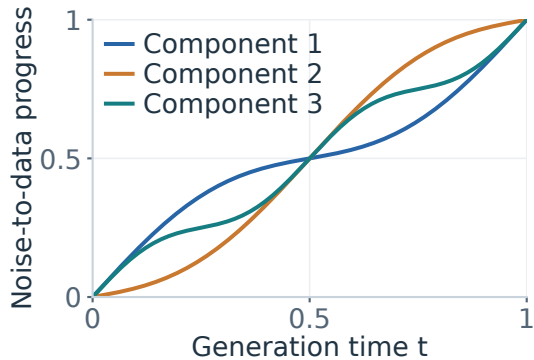
$$D_{j \rightarrow i} = L_i(\text{noisier } j) - L_i(\text{cleaner } j)$$

Positive D: cleaner j helps denoise i.

Repeat over noise levels and contexts.

B. Schedules and module activity

Illustrative; not measured



Noise vector → active modules

JOINT DESIGN

Optimize schedules and transformer jointly; condition module activity on all component noise levels, with sparse attention and shared or switched MLPs.